Homework 2

Due: Oct. 18

Submit your homework outside the instructor's office:
Room 324, Building 3, SIST

Rules

- Please work on your own. Discussion is permissible, but extremely similar submissions will be judged as plagiarism.
- Please show all intermediate steps: a correct solution without an explanation will get zero credit.
- Please submit on time. No late submission will be accepted.
- Please prepare your submission in English only. No Chinese submission will be accepted.
- Please retain 2 decimal places if rounding is required (unless otherwise specified).

熊闻野 Wenye Xiong

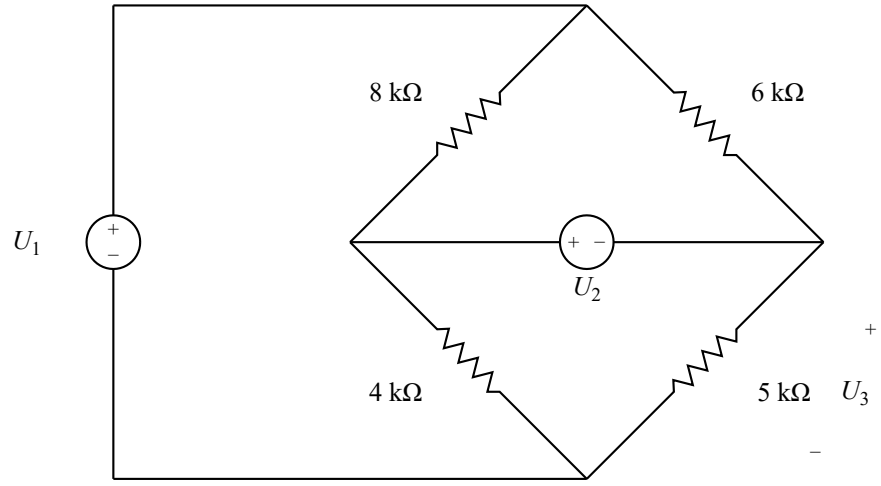
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1

For the circuit shown, use linearity and superposition to express U_3 as $U_3 = \alpha U_1 + \beta U_2$ and determine α and β.

(a) Find U_3 when $U_1 = 1.5\,\text{V}$ and $U_2 = 2\,\text{V}$.

(b) With U_1 unchanged, determine the value of U_2 that makes $U_3 = 0$.



(a) Set the bottom vertex of the bridge to 0 V, top vertex V_T, middle left V_L, middle Right V_R:

$V_T = U_1$, $V_L - V_R = U_2$, $V_R = U_3$

Contribution of U_1: short source U_2:

$R_{upp} = 8 \| 6 = \frac{24}{7}\,k\Omega$, $R_{low} = 4 \| 5 = \frac{20}{9}\,k\Omega$

$U_3^{(U_1)} = \frac{20/9}{24/7 + 20/9} U_1 = \frac{35}{89} U_1$

Contribution of U_2: short U_1:

$R_L = 8 \| 4 = \frac{8}{3}\,k\Omega$ $R_R = 6 \| 5 = \frac{30}{11}\,k\Omega$

$U_3^{(U_2)} = -\frac{R_R}{R_L + R_R} U_2 = -\frac{45}{89} U_2$

$U_3 = \frac{35}{89} U_1 - \frac{45}{89} U_2$, $\alpha = \frac{35}{89}$ $\beta = -\frac{45}{89}$

(a) $U_1 = 1.5\,V$, $U_2 = 2\,V$

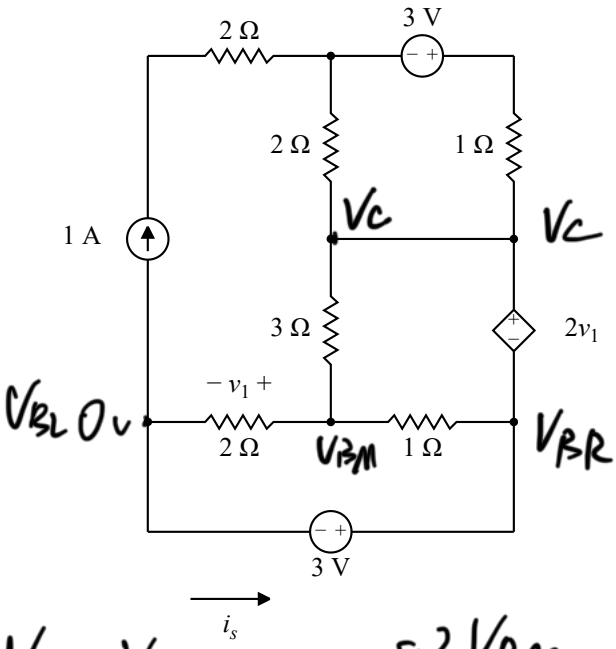
$U_3 = -\frac{75}{178}\,V = -0.42\,V$

(b) $U_3 = 0$, $U_1 = 1.5\,V$

$U_2 = \frac{7}{9} U_1 = \frac{7}{6}\,V = 1.17\,V$

2

(a) Use superposition to find the current i_s through the lower 3 V voltage source in the indicated direction. Show the separate contributions of the 1 A source and the two 3 V sources. Keep the dependent source active in each case.



$i_s = i_s' + i_s'' + i_s'''$

$= -1 + \frac{12}{7}$

$\approx 0.71\,A$

(a) $V_{BM} = V_1$ $\quad V_C - V_{BR} = 2V_1 = 2V_{BM}$

$i_s + I_{src} - \frac{V_{BM}}{2} = 0 \Rightarrow i_s = \frac{V_{BM}}{2} - I_{src}$ (KCL at BM)

Shorten 3V sources:

$V_{BR} = V_{BL} = 0V$, $\quad V_C = V_{BR} + 2v_1' = 2V_{BM}'$

KCL at node BM:

$\frac{V_{BM}' - 0}{2} + \frac{V_{BM}' - V_C}{3} + \frac{V_{BM}' - 0}{1} = 0 \Rightarrow V_{BM}' = 0V$

$i_s' = \frac{V_{BM}'}{2} - 1\,A = 0 - 1\,A = -1\,A$

Shorten the I_{src}, lower 3V source

KCL at node BM:

$\frac{V_{BM}''}{2} + \frac{V_{BM}'' - 2V_{BM}''}{3} + V_{BM}'' = 0 \Rightarrow V_{BM}'' = 0V$

$i_s'' = \frac{V_{BM}''}{2} - 0 = 0A$

$V_{BM}''' = \frac{24}{7}V \approx 3.43V$

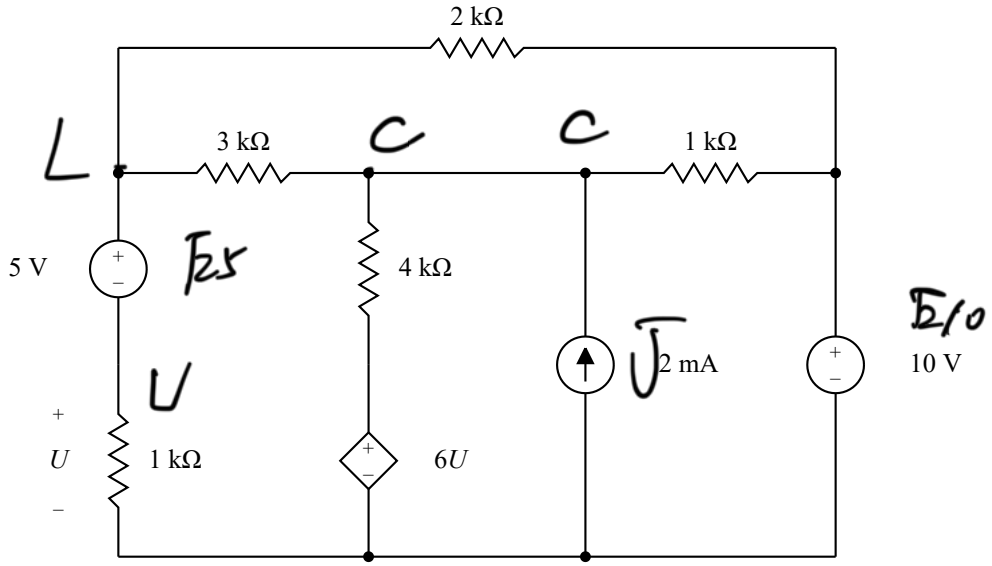
$i_s''' = \frac{12}{7}A = 1.71A$

Shorten I_{src}, upper 3V:

$\frac{V_{BM}''' - 0}{2} + \frac{V_{BM}''' - (3 + 2V_{BM}''')}{3} + \frac{V_{BM}''' - 3}{1} = 0$

3

Use superposition to find the voltage U across the 1 kΩ resistor with the indicated polarity. Show the separate contributions of the 5 V, 10 V, and 2 mA sources, keeping the dependent source active in each case.



$L = U + E_5$

KCL for left supernode

$$\frac{L-C}{3} + \frac{L-E_{10}}{2} + U = 0$$

$$11U - 2C = 3E_{10} - 5E_5$$

KCL for center node:

$$\frac{C-L}{3} + (C - E_{10}) + \frac{C-6U}{4} = J$$

$$19C - 22U = 4E_5 + 12E_{10} + 12J$$

$$U = \frac{-29E_5 + 27E_{10} + 8J}{55}$$

Individual contributions:

$$U^{(5V)} = \frac{-29 \cdot 5}{55} = -\frac{29}{11} V = -2.64V$$

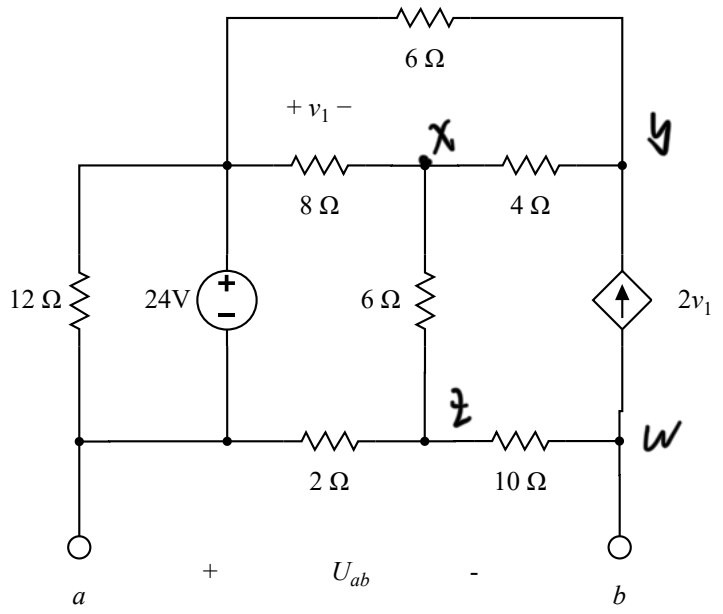
$$U^{(10V)} = \frac{27 \cdot 10}{55} = \frac{54}{11} V = 4.91 V$$

$$U^{(2mA)} = \frac{8 \cdot 2}{55} = \frac{16}{55} V = 0.29 V$$

$$U = \frac{141}{55} V = 2.56 V$$

4

Find the Thevenin equivalent circuit between terminals a and b.



$V_1 = 24V - x$

KCL:

$$\begin{cases} \frac{x-24}{8} + \frac{x-y}{4} + \frac{x-z}{6} = 0 \\ \frac{y-24}{6} + \frac{y-x}{4} - 2(24-x) = 0 \\ \frac{z}{2} + \frac{z-x}{6} + \frac{z-w}{10} = 0 \\ \frac{w-z}{10} + 2(24-x) = 0 \end{cases}$$

$$\Rightarrow \begin{cases} x = \frac{148}{7} V \\ y = 36 V \\ z = -\frac{23}{7} V \\ w = -\frac{423}{7} V \end{cases}$$

$V_{TH} = V_a - V_b$

$= 0 + \frac{423}{7} V$

$\approx 60.43 V$

Short the 24V source

$V_1 = -x$

Apply a test voltage V_t at b.

$R_{TH} = \frac{V_T}{I_t}$

$\frac{x}{8} + \frac{x-y}{4} + \frac{x-z}{6} = 0$

$\frac{y}{6} + \frac{y-x}{4} + 2x = 0$

$\frac{z}{2} + \frac{z-x}{6} + \frac{z-V_t}{10} = 0$

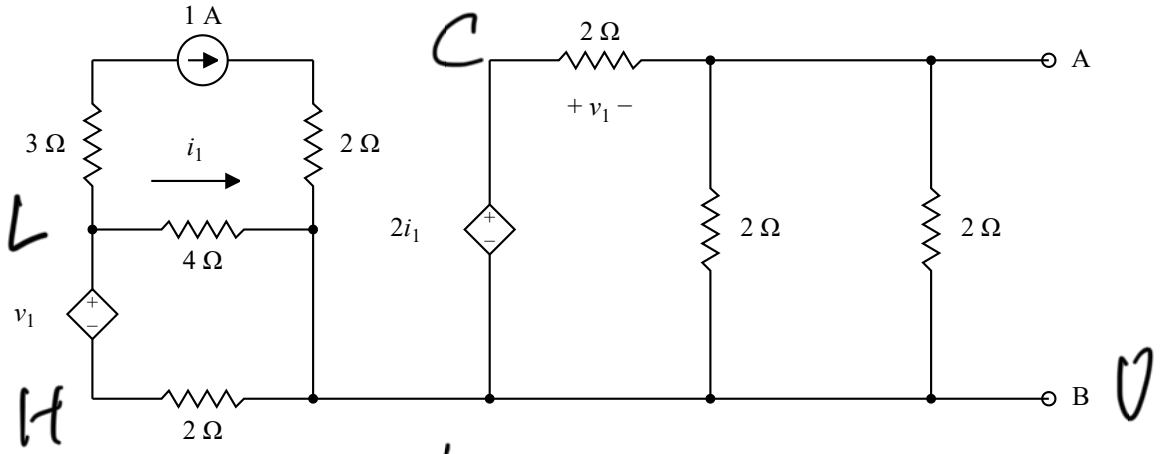
$\Rightarrow x = \frac{20}{1431} V_t \quad z = \frac{191}{1431} V_t$

$I_t = \frac{V_t - z}{10} - 2x = \frac{28}{477} V_t$

$R_{TH} = \frac{477}{28} \Omega \approx 17.04 \Omega$

5

Find the Thevenin equivalent circuit seen at terminals A and B. State the open-circuit voltage $V_{\text{Th}} = V_A - V_B$ and the equivalent resistance R_{Th}, and draw the equivalent circuit. The controlling voltage v_1 and current i_1 are defned by the polarity and direction shown.



$i_1 = \frac{L}{4}$, $C = 2i_1 = \frac{L}{2}$. $V_1 = C - V_A$ $L - H = V_1$

KCL for the left supernode (L, H):

$$\frac{L}{4} + \frac{H}{2} + 1 = 0 \Rightarrow 3L - 2V_1 = -4 \Rightarrow L = -2 - V_A$$

KCL at open terminal A:

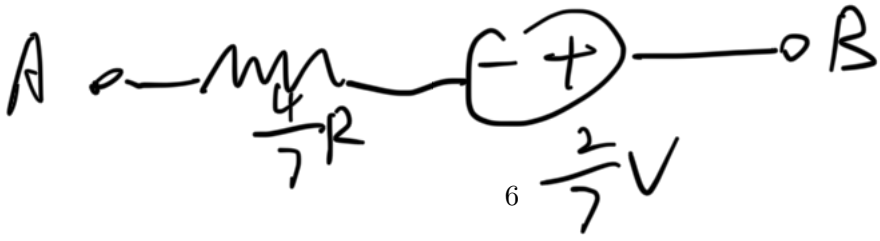
$$\frac{V_A - C}{2} + \frac{V_A}{2} + \frac{V_A}{2} = 0 \Rightarrow 3V_A = C = \frac{L}{2}$$

$\Rightarrow L = 6V_A$. $7V_A = -2$. $V_{Th} = -\frac{2}{7}V = -0.29V$

$3L - 2V_1 = 0$. $V_1 = \frac{L}{2} - V_t$. $I_t = \frac{V_t - C}{2} + \frac{V_t}{2} + \frac{V_t}{2}$

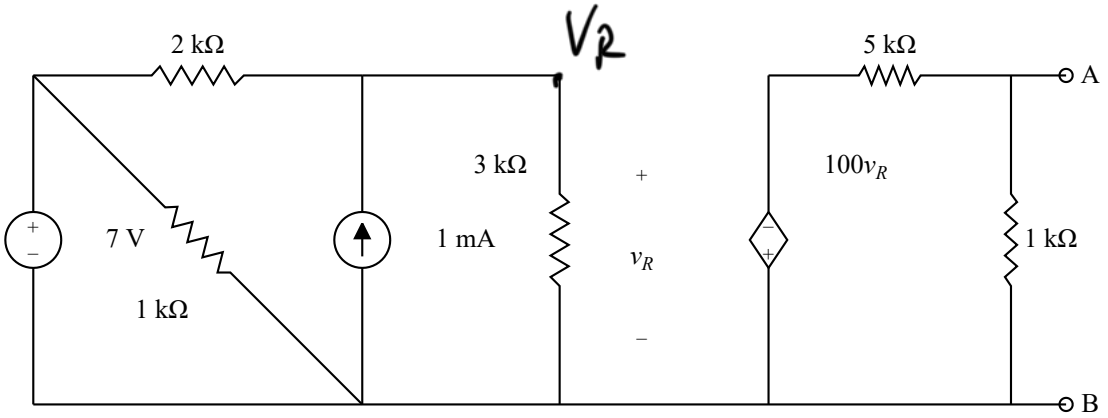
$= \frac{7}{4}V_t$

$R_{Th} = \frac{4}{7}\Omega = 0.57\Omega$



6

Find the Thevenin equivalent circuit and the Norton equivalent circuit at terminals A and B of the circuit below.



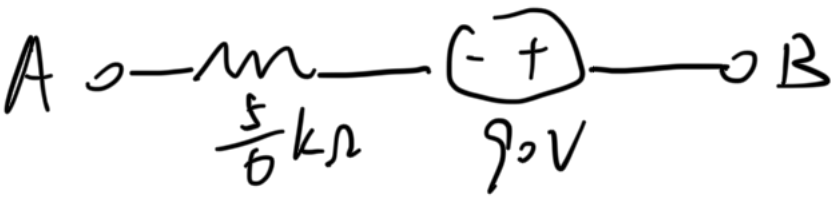
$\frac{V_R - 7}{2} + \frac{V_R}{3} = 1 \Rightarrow V_R = \frac{27}{5} V = 5.40V$

The dependent voltage source top voltage to be C:

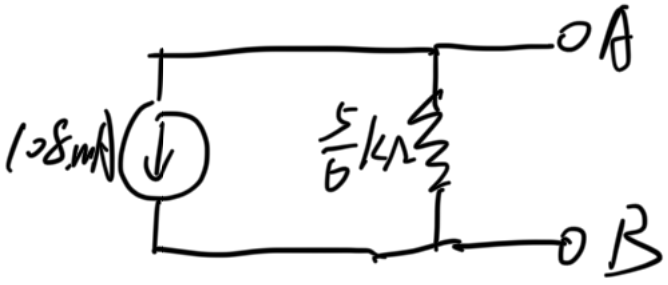
$0 - C = 100 V_R \Rightarrow C = -540V$

$V_{Th} = V_A = \frac{1}{1+5}(-540) = -90V$

$R_{Th} = R_N = 5 \| 1 k\Omega = \frac{5}{6} k\Omega = 833.33\Omega$

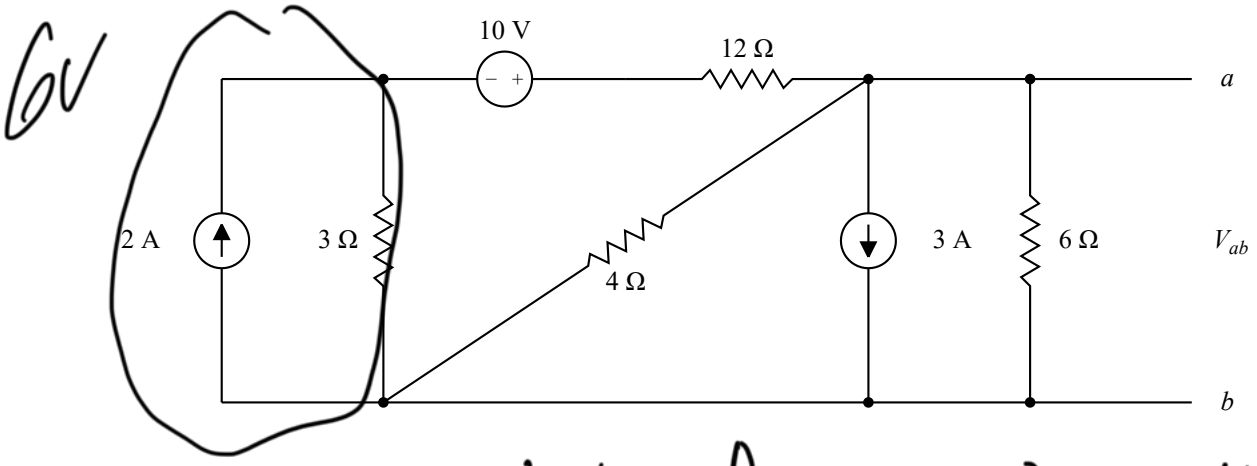


$I_N = \frac{90}{\frac{5}{6} \times 1000} = 0.108A = 108.00mA$



7

Use source transformation to find V_{ab} in the circuit below. Then find the Norton equivalent circuit at terminals a and b.



6V

$V_{Left,eq} = 6 + 10 = 16\,V$ $R_{Left,eq} = 3 + 12 = 15\,\Omega$

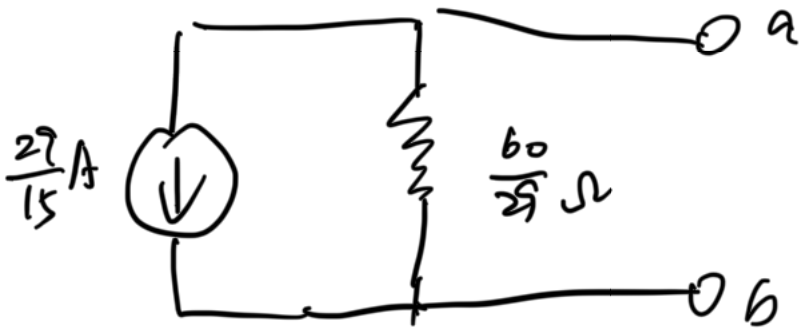
$I_{Left,eq} = \frac{16}{15}\,A$ $I_N = 3 - \frac{16}{15} = \frac{29}{15}\,A$

$R_N = 15 \parallel 4 \parallel 6 = \frac{60}{29}\,\Omega$

$V_{ab} = -I_N \cdot R_N = -4V$

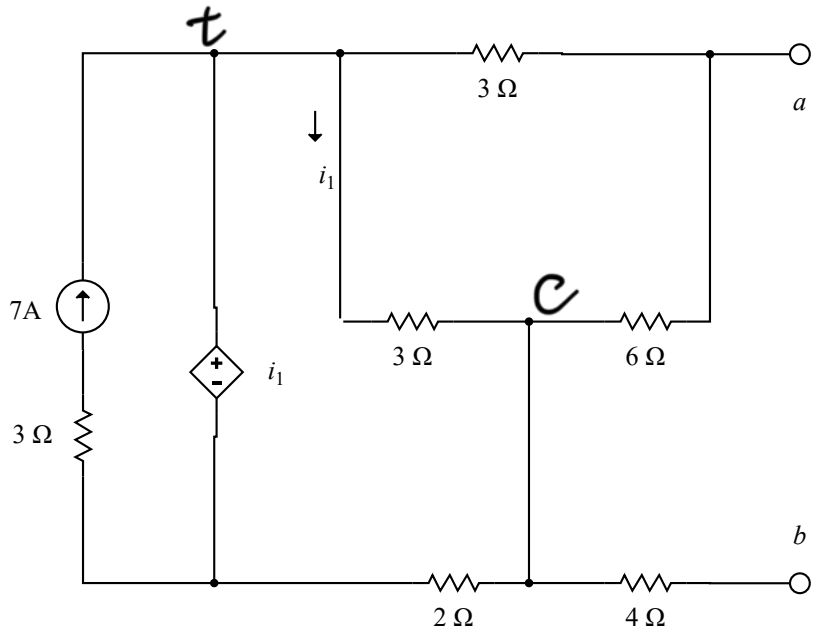
$I_N = \frac{29}{15}\,A = 1.93A$ from a to b

$R_N = \frac{60}{29}\,\Omega = 2.07\,\Omega$



8

Draw the Norton equivalent circuit between terminals a and b



$i_1 = \frac{t-c}{3}$ $\quad t = i_1 \Rightarrow \quad c = -2t$

KCL at a: $\frac{c-t}{3} + \frac{c-V_a}{6} + \frac{c}{2} = 0$, $V_a = 0$

$\Rightarrow t = 0$, $V_{oc} = 0V$

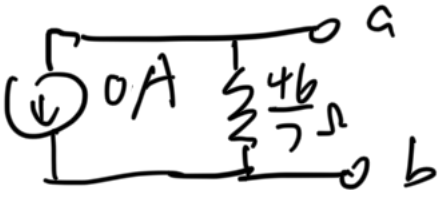
Open the independent 7A source, inject 1A at a, withdraw 1A at b. $\frac{V_a - t}{3} + \frac{V_a - c}{6} = 1$, $\frac{V_b - c}{4} = -1$

$\Rightarrow V_a = 2$, $V_b = c - 4$, KCL at c now gives:

$$\frac{c-t}{3} + \frac{c-V_a}{6} + \frac{c}{2} + \frac{c-V_b}{4} = 0$$

$$-\frac{7}{3}t + \frac{2}{3} = 0 \Rightarrow t = \frac{2}{7} \quad c = -\frac{4}{7} \quad V_b = -\frac{32}{7}$$

$R_N = \frac{V_a - V_b}{1A} = \frac{46}{7}\Omega \approx 6.57\Omega$ $\quad I_N = 0A$

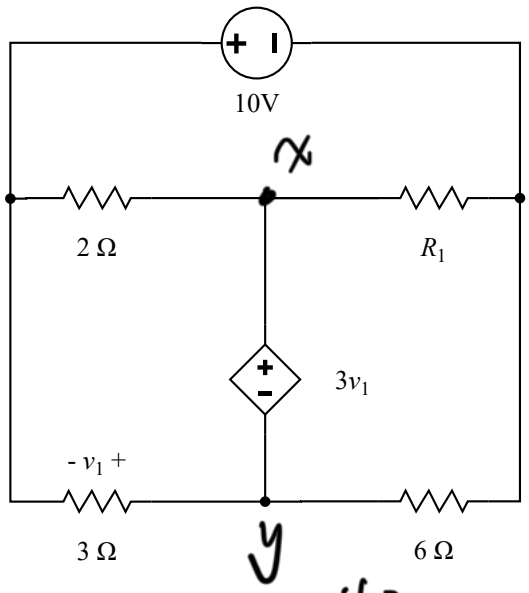


9

The resistance $R_1 \geq 0$ is adjustable. Keep every other element unchanged and answer the following two parts separately.

(a) Find the value of R_1 that maximizes the power absorbed by R_1. Calculate that maximum power.

(b) Find the value of R_1 that maximizes the power delivered by the 10 V source. Calculate that maximum delivered power.



$v_1 = y - 10$

$x - y = 3v_1$

$= 3(y-10)$

$\Rightarrow x = 4y - 30$

KCL for supernode (x, y):

$$\frac{x-10}{2} + I + \frac{y-10}{3} + \frac{y}{6} = 0$$

$$\Rightarrow x = \frac{22}{3} - \frac{8}{5} I$$

$V_{Th} = \frac{22}{3} V \quad . \quad R_{Th} = \frac{8}{5} \Omega$

(a) With $R = R_1 \geq 0$, $x = RI$

$$I = \frac{V_{Th}}{R + R_{Th}} \qquad P_R = \frac{V_{Th}^2 R}{(R+R_{Th})^2}$$

$$\frac{dP_R}{dR} = V_{Th}^2 \frac{R_{Th} - R}{(R+R_{Th})^3}$$

$R_1 = R_{Th} = \frac{8}{5} \Omega = 1.60 \Omega$ $\qquad P_{R_1 max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{605}{72} W = 8.40 W$

(b) $x = \frac{110R}{15R+24} \qquad y = \frac{x+30}{4} = \frac{140R+180}{15R+24}$

$R = R_1$

$$I_{10V} = \frac{10-x}{2} + \frac{10-y}{3}$$

$$= \frac{70(R+6)}{9(5R+8)}$$

$$P_{10V} = 10 \cdot I_{10V} = \frac{700(R+6)}{9(5R+8)}$$

$$\frac{dP_{10V}}{dR} = -\frac{15400}{9(5R+8)^2} < 0$$

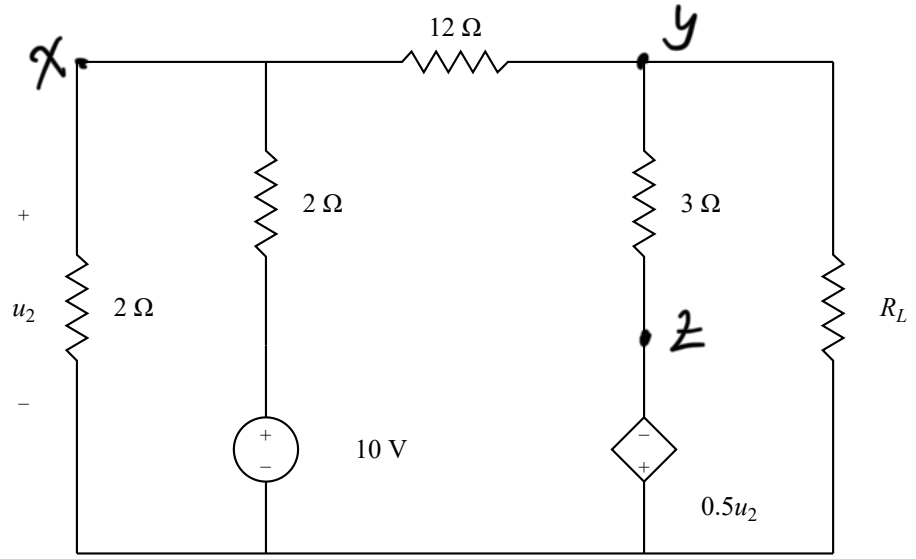
$R_1 = 0.00 \Omega$ maximum

$I_{10V} = \frac{35}{6} A$

$P_{10V,max} = 10\left(\frac{35}{6}\right) = 58.33 W$

10

The load resistance R_L can be adjusted. Find the value of R_L that receives maximum power and calculate that maximum power. At this operating point, find the ratio of the power delivered to R_L to the power supplied by the independent 10 V voltage source.



$z = -0.5u_2 = -\frac{x}{2}$

Remove R_L, KCL:

$$\frac{x}{2} + \frac{x-10}{2} + \frac{x-y}{12} = 0$$

$$\frac{y-x}{12} + \frac{y+x/2}{3} = 0$$

$$x+5y=0, \quad 13x-y=60$$

$$\Rightarrow V_{Th} = y = -\frac{10}{11}\,V$$

Short 10V:

$$\frac{x}{2} + \frac{x}{2} + \frac{x-y}{12} = 0$$

$$x = \frac{y}{13}$$

$$I_t = \frac{y-x}{12} + \frac{y+x/2}{3} = \frac{5y+x}{12} = \frac{11}{20}y$$

$$R_{Th} = \frac{26}{11}\,\Omega$$

$$R_L = R_{Th} = \frac{26}{11}\,\Omega = 2.36\,\Omega$$

$$P_{L,max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{25}{286}\,W = 87.4\,mW$$

$R_L = R_{Th}$, $y = \frac{V_{Th}}{2} = -\frac{5}{11}\,V$

$$x = \frac{60y}{13} = \frac{655}{143}\,V$$

$$I_{10V} = \frac{10-x}{2} = \frac{775}{286}\,A$$

$$P_{10V} = 10 \cdot I_{10V} = 27.10\,W$$

$$\frac{P_{L,max}}{P_{10}} = \frac{1}{310} \approx 0.32\%$$